TABLE I: Number of projects in our sample after applying the exclusion criteria

Year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
Projects	14,665	14,353	14,922	15,317	16,208	15,785	14,964	15,476	16,924	17,862

TABLE II: Differences in methodology between KOCHHARSTUDY [4] and our study

	Our study	**KOCHHARSTUDY [4]**
Time range	2012 to 2021	2012
Number of projects	20,000 per year	20,817
Project collection	Used WoC	Used GitHub API
Calculating LOC	Git commands	SLOCCount [17] utility
Counting bugs	Used bug referencing commits	Used GitHub issue tracking system
Definition of test case	Calculated test cases using @Test annotation	Considered test files same as test cases
Programming language	Java	Multiple programming languages

TABLE III: The trend (in %) of projects having test cases and test files over the past decade (2012-2021)

Year	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021
% of projects containing test files	48.37	45.99	44.65	55.38	57.87	59.97	62.87	61.17	60.43	58.64
% of projects containing test cases	29.06	27.58	24.81	28.24	42.65	47.60	50.92	50.70	50.31	48.96

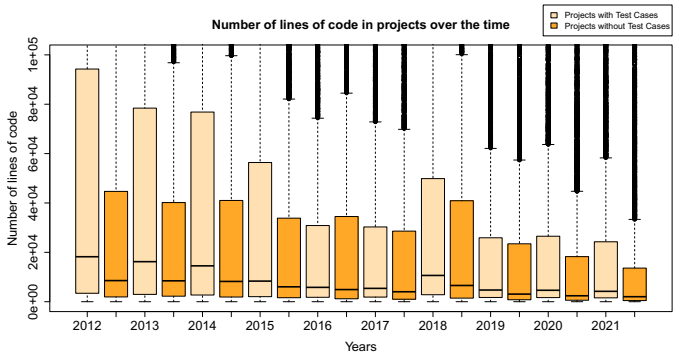


Fig. 1: Comparison of Lines of Code between Java Projects with and without Test Cases from 2012-2021

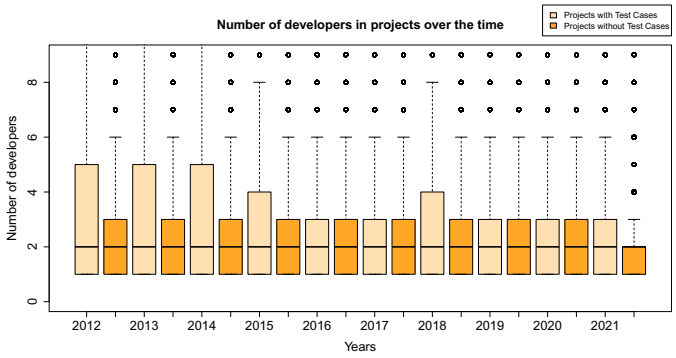


Fig. 2: Comparison of Number of Authors between Java Projects, with and without Test Cases, from 2012-2021

Additionally, KOCHHARSTUDY [4] used the MWW test to validate their conclusion and showed p-value < 2.2 e^{-16} [4]. In our study, for the MWW test, we observed p-values = 2.2 e^{-16} for all the years except 2016, when the p-value was 5.587 e^{-14}. This observation demonstrates a statistically significant difference between the LOC in projects with test cases and those without, which confirms the stability of KOCHHARSTUDY [4]'s conclusion from 2012 to 2021.

Furthermore, to find out how the project size and the number of test cases correlate, KOCHHARSTUDY [4] used Spearman's rho and reported $\rho = 0.427$ in their study [4]. In our case, the ρ values were between 0.26 and 0.37 (p-value = 2.2 e^{-16}) for all the years, which is similar to their reported value. Therefore, it is clear that their statement "there is a positive correlation between the number of test cases and the number of LOC" [4] is stable for all the years in our study.

KOCHHARSTUDY [4] further investigated the relationship between project size and the number of test cases per line of code. They observed a negative correlation in this scenario and reported Spearman's $\rho = -0.451$. We also observed negative rho values in the range of -0.60 to -0.68 with p-value = 2.2 e^{-16} (< α=0.05) in our study, which validates their finding that "the number of test cases per LOC decreases with increasing LOC" [4] for all years from 2012 to 2021.

> For the Java projects on GitHub from 2012 to 2021, the conclusions from KOCHHARSTUDY [4] that "Projects with test cases are bigger in size than projects without test cases." [4] and "the number of test cases per LOC decreases with increasing LOC" [4], remain stable through 2012-2021.

B. Correlation between Number of Developers and Test Cases (RQ2)

In their second research question, KOCHHARSTUDY [4] investigated, "Does the number of developers affect the number of test cases present in a project?" [4]. To re-investigate this, we counted the number of authors for each project in our sample for all the years. Figure 2 illustrates the number of authors in the Java projects.